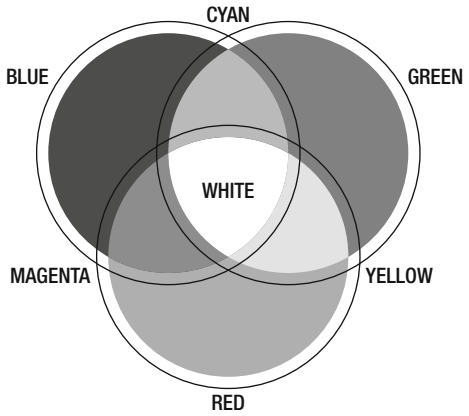

Figure 2-1. *RGB color wheel*

The particular LED we will be using is actually three separate LEDs—in the three primary colors of red, green, and blue—in one component package. Additionally, this LED is a common cathode type, meaning that the three LEDs share a common cathode pin, or connection to ground. This pin is the longest of the four pins coming out of the LED. We will connect each of the other three pins to the Arduino board through current limiting resistors, using hookup wires and a solderless breadboard.

To get an approximately equal or balanced output from each LED, we will use a 330 ohm resistor (ohm is a unit of measurement for resistance) connected to the red LED, or leg 1 of the RGB LED, and 220 ohm resistors to the other two, legs 3 and 4. If you don't have these resistors on hand, you can use the values 100 and 150 ohm or others that are close enough in value, anywhere between 100 ohms and 1 kilohm. If you use the same values for each color that's okay too, but each color's brightness might not be as closely matched. Some RGB LEDs are different from others so double-check the data sheet or product description as your mileage may vary. Refer to the following schematic and illustration for how to wire up this circuit.

Hooking It Up

For the projects in this book, we will follow a couple of diagrams that show in different ways how the hardware goes together. We begin with a **schematic**, like the one in Figure 2-2, which is basically a simple drawing that shows in the most direct way possible how to make a circuit with lines that connect simple, abstracted symbols for the electrical components. Where things get placed in a schematic is rarely where they end up in our actual physical circuit—it's just a way to show how one thing connects to another. Because schematics can be rather abstract, we will also include an illustration, like the one in Figure 2-3, showing one way that the finished circuit might look on the breadboard and how it connects to the Arduino board.

■ **Note** To find out more information on prototyping circuits, including how to read schematics and the symbols for different components, check out Chapter 12 later in this book.